Leo C. Stein

Contact Information

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Education

Ph.D., Physics, Massachusetts Institute of Technology, Cambridge, MA, USA **May 2012**

Dissertation Advisor: Prof. Scott Hughes

Dissertation Title: *Probes of strong-field gravity*

B.S., Physics, California Institute of Technology, Pasadena, CA, USA **June 2006**

Degree conferred with honor.

Senior Thesis Advisors: Dr. Patrick Sutton and Prof. Alan Weinstein

Employment

Associate Professor, University of Mississippi, Oxford, MS USA **July 2024–Present**

Assistant Professor, University of Mississippi, Oxford, MS USA **August 2018–June 2024**

Senior Postdoctoral Researcher, Caltech, Pasadena, CA USA **September 2015–August 2018**

NASA Einstein Fellow, Cornell, Ithaca NY, USA **September 2012–August 2015**

Research and Teaching Assistant, MIT, Cambridge MA, USA **September 2006–May 2012**

Teaching Assistant, Caltech, Pasadena, CA, USA **Fall 2004, Spring 2005**

Summer Research Fellow, Caltech, Pasadena, CA, USA **June–September 2003/2005**

Research Interests

General relativity (GR), gravitation, and astrophysical phenomena which can elucidate gravity. One major theme is pushing numerical and analytical gravitational-wave (GW) predictions to the precision frontier in advance of next-generation observatories. A second major theme is using GWs to test GR against beyond-GR models, in both theory-independent and theory-dependent models. This involves numerical relativity and renormalization methods applied to specific effective field models for beyond-GR theories.

Honors and Awards

Sloan Research Fellowship, Alfred P. Sloan Foundation, **2023–2025**

CAREER Award, NSF **2021–2026**

Einstein Postdoctoral Fellow, NASA **2012–2015**

Henry Kendall Teaching Award, Massachusetts Institute of Technology **2011**

Upperclass Merit Scholarship, California Institute of Technology **2005–2006**

Teaching Experience

Professor, University of Mississippi

Phys. 213, General physics I **Spring 2021**

Phys. 401, Electromagnetism I **Falls 2019–2022**

Phys. 402, Electromagnetism II **Springs 2019–2021**

Phys. 436, Intro to cosmology	**Fall 2023**
Phys. 463/4, Senior research project	**Fall 2020, Spring 2021, Fall 2023**
Phys. 503/630, Graduate reading course	**Spring 2019, Falls 2020–2021**
Phys. 709, Graduate classical dynamics I	**Fall 2018**
Phys. 721, Graduate electrodynamics I	**Springs 2022–2024**
Phys. 722, Graduate electrodynamics II	**Falls 2022–2024**
Phys. 735, General relativity	**Fall 2024**
Phys. 750, General relativity II	**Spring 2020**

Guest Lecturer, California Institute of Technology

Ph236, General relativity	**Fall 2017**
Ph237, Gravitational Waves	**Spring 2016**

Guest Lecturer, Massachusetts Institute of Technology

8.901, Graduate Astrophysics I	**Spring 2011**

Teaching Assistant, Massachusetts Institute of Technology

8.942, Cosmology	**Fall 2011**
8.901, Graduate Astrophysics I	**Spring 2011**
8.286, The Early Universe	**Fall 2009**

Teaching Assistant, California Institute of Technology

Ph 7, Nuclear and Quantum Physics Lab	**Spring 2005**
Ph 5, Analog Electronics for Physicists	**Fall 2004**

Mentoring/ Supervision

Postdoctoral researchers

Károly Csukás	**Fall 2021–Summer 2024**
José Tomás Gálvez Ghersi	**Fall 2019–Spring 2021**
Now faculty at Universidad de Ingeniería y Tecnología, Peru	

Graduate students

David Bronicki, University of Mississippi	**Fall 2019–Summer 2023**
Subhayu Bagchi, University of Mississippi	**Fall 2019–present**
Aniket Khairnar, University of Mississippi	**Fall 2019–present**
Akshay Khadse, University of Mississippi	**Fall 2018–Summer 2024**
Lorena Magaña Zertuche, University of Mississippi	**Fall 2018–Summer 2024**
Now a postdoc at NBI, Copenhagen, Denmark	
Joe Rivest, University of Mississippi	**Fall 2018–Summer 2024**
Sashwat Tanay, University of Mississippi	**Fall 2018–Summer 2022**
Now a postdoc at LUTH, Meudon, France	
Maria (Masha) Okounkova, Caltech	**Fall 2015–Summer 2019**
Now faculty at Pasadena City College	
Baoyi Chen, Caltech	**Fall 2016–Summer 2018**

Undergraduate students

Wayne Zhao, Harvard **Summer 2016**

Now a graduate student at Princeton

Professional Activities, Outreach, and Service

LISA Consortium, Full member **2020–Present**

UMiss LISA Group leader **2020–Present**

Simulating eXtreme Spacetimes collaboration **2015–Present**

Executive committee member **2018–Present**

American Physical Society, member **2010–Present**

Division of Gravitational Physics

Secretary/Treasurer **2023–2026**

Executive Committee Member-at-Large **2016–2019**

Division of Astrophysics

Conference organizer

Simulating Extreme Spacetimes with SpEC and SpECTRE, ICERM **August 2024**
Week-long international workshop, ∼85 participants

New frontiers in strong gravity, Benasque, Spain **July 2024**
Two week international conference, ∼70 participants

Nonlinear Aspects of General Relativity, Princeton PCTS **October 2023**
Four day workshop, ∼100 participants

Numerical Relativity Community Summer School, ICERM **August 2022**
Week-long international summer school, 150 participants

New frontiers in strong gravity, Benasque, Spain **July 2022**
Two week international conference, 100 participants

Numerical Relativity beyond General Relativity, Benasque, Spain **June 2018**
Week-long international workshop, 59 participants

34th Pacific Coast Gravity Meeting (PCGM), Caltech **March 2018**
Two day conference, ∼ 125 participants

Unifying Tests of General Relativity, Caltech **July 2016**
Three day workshop, 52 participants

Seminar organizer

TAPIR seminar, Caltech **Fall 2015–Spring 2018**

General Relativity Informal Tea-Time Series (GRITTS), MIT **Fall 2011–Spring 2012**

MKI Journal Club, MIT **Fall 2007–Spring 2010**

Conference session chair; Judge for best student speaker award

April APS meeting, NY, NY **April 2022**

Midwest relativity meeting, Grand Rapids, MI **October 2019**

April APS meeting, Columbus, OH **April 2018**

34th Pacific Coast Gravity Meeting (PCGM), Caltech **March 2018**

33rd Pacific Coast Gravity Meeting (PCGM), UCSB **March 2017**

"April" APS meeting, Washington D.C. **January 2017**

32nd Pacific Coast Gravity Meeting (PCGM), CSU Fullerton **April 2016**

Theoretical Astrophysics in Southern California (TASC), CSU Fullerton **November 2015**

Journal referee

American Journal of Physics, Classical and Quantum Gravity, Journal of Cosmology and Astroparticle Physics, Journal of Open Source Software, General Relativity and Gravitation, Monthly Notices of the Royal Astronomical Society, Physics Letters B, Physical Review D, Physical Review Letters, Physical Review X, Reviews of Modern Physics, The Astrophysical Journal Letters, The Physics Teacher

Agency work

Reviewer for NSF, NASA

Outreach

Oxford Science Café **April 2019**
Lecture: "The truth about black holes"

Guest on the *Starts With a Bang* podcast **March 25, 2019**
Episode 42: Black holes and gravitation

Invited speaker for Latin American Webinar on Physics **March 13, 2019**
Webinar 75: "Testing Einstein with numerical relativity"

Caltech astronomy public lecture series speaker **March 2018**
Lecture: "The truth about black holes"

Astronomy on Tap public lecture series speaker and volunteer **2016–2018**
Close to a monthly basis

Caltech astronomy public lecture series panelist and emcee **2016–2018**
Approximately every three months

Invited guest lecture on black holes and gravitational waves **November 2017**
Science of Space and Time, Hampshire College

Invited video Q&A session, public high school physics class **June 2017**
The Nova Project school, Seattle

Guest on *The Titanium Physicists Podcast*
Episode 80: Picturing the Bach Hole **August 21, 2019**
Episode 64: The edges of Einstein **April 25, 2016**
Episode 62: Black Bells **February 1, 2016**

Quora Q&A Session on gravitational waves and first detection **February 17, 2016**
83.9k+ views, 20.8k+ followers

Invited guest host, public screening of *COSMOS* with Q&A, **March/June 2014**
Science Cabaret/Cornell

Invited public talk at *Frontiers of Cornell Astronomy*, **November 2013**
Cornell Friends of Astronomy

Invited video chat, *Topics in Physics* course, **July 2013**
Stanford Education Program for Gifted Youth

Computer Skills

Expert in Mathematica, C/C++, Python, Bash. Proficient in Javascript. Experience in Haskell, Java, Julia. Expert at *nix and HPC. Markup languages: LaTeX, HTML, CSS, Markdown.

Software—Most contributions can be found at `https://github.com/duetosymmetry`. Member of the *Simulating eXtreme Spacetimes* (SXS) collaboration, contributor to the Spectral Einstein Code (SpEC). Member of the *Black Hole Perturbation Toolkit*. Author of `qnm` python package (`https://github.com/duetosymmetry/qnm`). Core collaborator on xAct (`http://xact.es`) abstract tensor calculus package for Mathematica. Coauthor of xTerior package for exterior differential geometry under xAct. Co-maintainer of community contributions at `http://contrib.xact.es`. Developed arXiv-keys browser extension/add-on for Chrome/Firefox. Author of `orcidlink` and coauthor of `gridpapers` packages for LaTeX.

Publication Summary

h-index —As of 2024-10-13: 62 (according to Google Scholar), or 55 (according to INSPIRE). Both include collaboration papers.

Top five cited —Excluding LIGO/Virgo collaboration papers.

1. Berti, E., (5 authors), **Stein, L. C.**, (46 more authors) (2015) *Testing General Relativity with Present and Future Astrophysical Observations*, Class. Quantum Grav. **32** 243001 [arXiv:1501.07274].
2. Barack, L., *et al.* (2019) *Black holes, gravitational waves and fundamental physics: a roadmap*, Class. Quantum Grav. **36** 143001 [arXiv:1806.05195].
3. Boyle, M., *et al.* (**LCS** is corresponding author) (2019) *The SXS Collaboration catalog of binary black hole simulations*, Class. Quantum Grav. **36** 195006 [arXiv:1904.04831].
4. Varma, V, *et al.* (2019) *Surrogate models for precessing binary black hole simulations with unequal masses*, Phys. Rev. Research **1**, 033015 [arXiv:1905.09300].
5. Yunes, N., **Stein, L. C.** (2011), *Nonspinning black holes in alternative theories of gravity*, Phys. Rev. D **83** 104002 [arXiv:1101.2921].

Submitted Publications

65. De Amicis, M. (5 authors), **Stein, L. C.**, (13 more authors) (2024) *Late-time tails in nonlinear evolutions of merging black holes*, [arXiv:2412.06887].
64. Witzany, V. Skoupý, V., **Stein, L. C.**, Tanay, S., (2024) *Actions of spinning compact binaries: Spinning particle in Kerr matched to dynamics at 1.5 post-Newtonian order*, [arXiv:2411.09742].
63. Khairnar, A., **Stein, L. C.**, Boyle, M., (2024) *Approximate helical symmetry in compact binaries*, [arXiv:2410.16373].
62. Magaña Zertuche, L., **Stein, L. C.**, *et al.*, (2024) *High-Precision Ringdown Surrogate Model for Non-Precessing Binary Black Holes*, [arXiv:2408.05300].
61. Zhu, H., (9 authors), **Stein, L. C.**, (2024) *Imprints of Changing Mass and Spin on Black Hole Ringdown*, [arXiv:2404.12424].

Collaboration Publications

From 2008–2012, I was coauthor on 34 refereed LIGO and/or LIGO/Virgo collaboration publications. I only list short author-list publications below.

Refereed Publications

60. Sun, D., Boyle, M., Mitman, K., Scheel, M. A., **Stein, L. C.**, Teukolsky, S. A., Varma, V., (2024) *Optimizing post-Newtonian parameters and fixing the BMS frame for numerical-relativity waveform hybridizations*, Phys. Rev. D. **110** 104076, [arXiv:2403.10278].
59. Mitman, K., Boyle, M., **Stein, L. C.**, *et al.*, (2024) *A Review of Gravitational Memory and BMS Frame Fixing in Numerical Relativity*, Class. Quantum Grav. **41** 223001, [arXiv:2405.08868].
58. **Stein, L. C.**, (2024) *Can a radiation gauge be horizon-locking?*, Class. Quantum Grav. **41** 157001 [arXiv:2404.10113].

57. Samanta, R., Tanay, S., **Stein, L. C.**, (2023) *Closed-form solutions of spinning, eccentric binary black holes at 1.5 post-Newtonian order*, Phys. Rev. D **108**, 124039 [arXiv:2210.01605].
56. Bronicki, D., Cárdenas-Avendaño, A., **Stein, L. C.**, (2023) *Tidally-induced nonlinear resonances in EMRIs with an analogue model*, Class. Quantum Grav. **40** 215015 [arXiv:2203.08841].
55. Yoo, J., *et al.*, (2023) *Numerical relativity surrogate model with memory effects and post-Newtonian hybridization*, Phys. Rev. D **108**, 064027 [arXiv:2306.03148].
54. Ma, S., Varma, V., **Stein, L. C.**, *et al.* (2023) *Numerical simulations of black hole–neutron star mergers in scalar-tensor gravity*, Phys. Rev. D **107**, 124051 [arXiv:2304.11836].
53. Tanay, S., **Stein, L. C.**, Cho, G., (2023) *Action-angle variables of a binary black-hole with arbitrary eccentricity, spins, and masses at 1.5 post-Newtonian order*, Phys. Rev. D **107**, 103040 [arXiv:2110.15351].
52. Grant, A. M., Saffer, A., **Stein, L. C.**, Tahura, A., (2023) *Gravitational-wave energy and other fluxes in ghost-free bigravity*, Phys. Rev. D **107**, 044041 [arXiv:2208.02123].
51. Mitman, K., Lagos, M., **Stein, L. C.**, *et al.* (2023) *Nonlinearities in black hole ringdowns*, Phys. Rev. Lett. **130**, 081402 [arXiv:2208.07380]. Editors' Suggestion, Featured in Physics.
50. Clark, W. A., Gomes, M. W., Rodriguez-Gonzalez, A., **Stein, L. C.**, Strogatz, S. H., (2023) *Surprises in a classic boundary-layer problem*, SIAM Review 2023 65:1, 291-315 [arXiv:2107.11624].
49. Mitman, K., **Stein, L. C.**, Boyle, M., *et al.* (2022) *Fixing the BMS Frame of Numerical Relativity Waveforms with BMS Charges*, Phys. Rev. D **106**, 084029 [arXiv:2208.04356].
48. Okounkova, M, Farr, W. M., Isi, M., **Stein, L. C.**, (2022) *Constraining gravitational wave amplitude birefringence and Chern-Simons gravity with GWTC-2*, Phys. Rev. D **106**, 044067 [arXiv:2101.11153].
47. Magaña Zertuche, L., Mitman, K., Khera, N., **Stein, L. C.**, et al., (2022) *High Precision Ringdown Modeling: Multimode Fits and BMS Frames*, Phys. Rev. D **105**, 104015 [arXiv:2110.15922].
46. Gálvez Ghersi, J. T., **Stein, L. C.**, (2021) *Numerical renormalization group-based approach to secular perturbation theory*, Phys. Rev. E **104**, 034219 [arXiv:2106.08410].
45. Mitman, K., Khera, N., Iozzo, D. A. B., **Stein, L. C.**, et al., (2021) *Fixing the BMS frame of numerical relativity waveforms*, Phys. Rev. D **104**, 024051 [arXiv:2105.02300].
44. Iozzo, D. A. B., Khera, N., **Stein, L. C.**, et al., (2021) *Comparing Remnant Properties from Horizon Data and Asymptotic Data in Numerical Relativity*, Phys. Rev. D **103**, 124029 [arXiv:2104.07052].
43. Tahura, S., Nichols, D. A., Saffer, A., **Stein, L. C.**, Yagi, K. (2020) *Brans-Dicke theory in Bondi-Sachs form: Asymptotically flat solutions, asymptotic symmetries and gravitational-wave memory effects*, Phys. Rev. D **103**, 104026 [arXiv:2007.13799].
42. Tanay, S., **Stein, L. C.**, Gálvez Ghersi, J. T., (2020) *Integrability of eccentric, spinning black hole binaries up to second post-Newtonian order*, Phys. Rev. D **103**, 064066 [arXiv:2012.06586].
41. Gálvez Ghersi, J. T., **Stein, L. C.**, (2020) *A fixed point for black hole distributions*, Class. Quantum Grav. **38** 045012 [arXiv:2007.11578].
40. Okounkova, M., **Stein, L. C.**, Moxon, J., Scheel, M. A., Teukolsky, S. A., (2020) *Numerical relativity simulation of GW150914 beyond general relativity*, Phys. Rev. D **101**, 104016 [arXiv:1911.02588].
39. **Stein, L. C.**, Warburton, N., (2020) *Location of the last stable orbit in Kerr spacetime*, Phys. Rev. D **101**, 064007 [arXiv:1912.07609].
38. Okounkova, M., **Stein, L. C.**, Scheel, M. A., Teukolsky, S. A., (2019) *Numerical binary black hole collisions in dynamical Chern-Simons gravity*, Phys. Rev. D **100**, 104026 [arXiv:1906.08789].
37. Varma, V, *et al.* (2019) *Surrogate models for precessing binary black hole simulations with unequal masses*, Phys. Rev. Research **1**, 033015 [arXiv:1905.09300].

36. **Stein, L. C.**, (2019) `qnm`: *A Python package for calculating Kerr quasinormal modes, separation constants, and spherical-spheroidal mixing coefficients*, J. Open Source Softw., 4(42), 1683 [arXiv:1908.10377].
35. Boyle, M., *et al.* (**LCS** is corresponding author) (2019) *The SXS Collaboration catalog of binary black hole simulations*, Class. Quantum Grav. **36** 195006 [arXiv:1904.04831].
34. Barack, L., *et al.* (2019) *Black holes, gravitational waves and fundamental physics: a roadmap*, Class. Quantum Grav. **36** 143001 [arXiv:1806.05195].
33. Varma, V., **Stein, L. C.**, Gerosa, D., (2019) *The binary black hole explorer: on-the-fly visualizations of precessing binary black holes*, Class. Quantum Grav. **36** 095007 [arXiv:1811.06552], [project website].
32. Varma, V., Gerosa, D., **Stein, L. C.**, Hébert, F., Zhang, H., (2019) *High-accuracy mass, spin, and recoil predictions of generic black-hole merger remnants*, Phys. Rev. Lett. **122**, 011101 [arXiv:1809.09125].
31. Isi, M., **Stein, L. C.** (2018) *Measuring stochastic gravitational-wave energy beyond general relativity*, Phys. Rev. D **98**, 104025 [arXiv:1807.02123].
30. Prabhu, K., **Stein, L. C.** (2018) *Black hole scalar charge from a topological horizon integral in Einstein-dilaton-Gauss-Bonnet gravity*, Phys. Rev. D **98**, 021503(R) (Rapid Communication) [arXiv:1805.02668].
29. Gerosa, D., Hébert, F., **Stein, L. C.** (2018) *Black-hole kicks from numerical-relativity surrogate models*, Phys. Rev. D **97**, 104049 [arXiv:1802.04276].
28. Chen, B., **Stein, L. C.** (2018) *Deformation of extremal black holes from stringy interactions*, Phys. Rev. D **97**, 084012 [arXiv:1802.02159].
27. Chen, B., **Stein, L. C.** (2017) *Separating metric perturbations in near-horizon extremal Kerr*, Phys. Rev. D **96**, 064017 [arXiv:1707.05319].
26. Okounkova, M., **Stein, L. C.**, Scheel, M. A., Hemberger, D. A. (2017) *Numerical binary black hole mergers in dynamical Chern-Simons: I. Scalar field*, Phys. Rev. D **96**, 044020 [arXiv:1705.07924].
25. Tso, R., Isi, M., Chen, Y., **Stein, L. C.** (2017) *Modeling the Dispersion and Polarization Content of Gravitational Waves for Tests of General Relativity*, CPT and Lorentz Symmetry: pp. 205–208 [arXiv:1608.01284].
24. McNees, R., **Stein, L. C.**, Yunes, N. (2016) *Extremal Black Holes in Dynamical Chern-Simons Gravity*, Class. Quantum Grav. **33** 235013 [arXiv:1512.05453].
23. Flanagan, É. É., Nichols, D. A., **Stein, L. C.**, Vines, J. (2016) *Prescriptions for Measuring and Transporting Local Angular Momenta in General Relativity*, Phys. Rev. D **93**, 104007 [arXiv:1602.01847].
22. Yagi, K., **Stein, L. C.** (2016) *Black Hole Based Tests of General Relativity*, Class. Quantum Grav. **33** 054001 [arXiv:1602.02413].
21. Yagi, K., **Stein, L. C.**, Yunes, N. (2016) *Challenging the Presence of Scalar Charge and Dipolar Radiation in Binary Pulsars*, Phys. Rev. D **93** 024010 [arXiv:1510.02152].
20. Berti, E., (5 authors), **Stein, L. C.**, (46 more authors) (2015) *Testing General Relativity with Present and Future Astrophysical Observations*, Class. Quantum Grav. **32** 243001 [arXiv:1501.07274].
19. Tsang, D., Galley, C. R., **Stein, L. C.**, Turner, A. (2015) *"Slimplectic" Integrators: Variational Integrators for General Nonconservative Systems*, ApJ **809** L9 [arXiv:1506.08443].
18. Yagi, K., **Stein, L. C.**, Pappas, G., Yunes, N., Apostolatos, T. (2014) *Why I-Love-Q: Explaining why universality emerges in compact objects*, Phys. Rev. D **90** 063010 [arXiv:1406.7587].

17. **Stein, L. C.** (2014) *Rapidly rotating black holes in dynamical Chern-Simons gravity: Decoupling limit solutions and breakdown*, Phys. Rev. D **90** 044061 [arXiv:1407.2350].
16. **Stein, L. C.**, Yagi, K., Yunes, N. (2014) *Three-Hair Newtonian Relations for Rotating Stars*, ApJ **788** 15 [arXiv:1312.4532].
15. **Stein, L. C.**, Yagi, K. (2014) *Parameterizing and constraining scalar corrections to general relativity*, Phys. Rev. D **89** 044026 [arXiv:1310.6743].
14. Yagi, K., **Stein, L. C.**, Yunes, N., Tanaka, T. (2013) *Isolated and Binary Neutron Stars in Dynamical Chern-Simons Gravity*, Phys. Rev. D **87** 084058 [arXiv:1302.1918].
13. Yagi, K., **Stein, L. C.**, Yunes, N., Tanaka, T. (2012), *Post-Newtonian, Quasi-Circular Binary Inspirals in Quadratic Modified Gravity*, Phys. Rev. D **85** 064022 [arXiv:1110.5950].
12. Vigeland, S., Yunes, N., **Stein, L. C.** (2011), *Bumpy black holes in alternative theories of gravity*, Phys. Rev. D **83** 104027 [arXiv:1102.3706].
11. Yunes, N., **Stein, L. C.** (2011), *Nonspinning black holes in alternative theories of gravity*, Phys. Rev. D **83** 104002 [arXiv:1101.2921].
10. **Stein, L. C.**, Yunes, N. (2011), *Effective gravitational wave stress-energy tensor in alternative theories of gravity*, Phys. Rev. D **83** 064038 [arXiv:1012.3144].
9. Lutomirski, A., Tegmark, M., Sanchez, N. J., **Stein, L. C.**, Urry, W. L., Zaldarriaga, M. (2011), *Solving the corner-turning problem for large interferometers*, MNRAS **410** 2075 [arXiv:0910.1351].
8. Sutton, P., Jones, G., Chatterji, S., Kalmus, P., Leonor, I., Poprocki, S., Rollins, J., Searle, A., **Stein, L.**, Tinto, M., Was, M. (2010), *X-Pipeline: an analysis package for autonomous gravitational-wave burst searches*, New J. Phys. **12** 053034 [arXiv:0908.3665].
7. Chatterji, S., Lazzarini, A., **Stein, L.**, Sutton, P., Searle, A. (2006), *Coherent network analysis technique for discriminating gravitational-wave bursts from instrumental noise*, Phys. Rev. D **74** 082005 [arXiv:gr-qc/0605002].

Unrefereed Publications

6. Galley, C. R., Tsang, D., **Stein, L. C.** (2014) *The principle of stationary nonconservative action for classical mechanics and field theories*, [arXiv:1412.3082].
5. **Stein, L. C.** (2014), *Note on Legendre decomposition of the Pontryagin density in Kerr*, [arXiv:1407.0744].
4. **Stein, L. C.** (2012), *Probes of Strong-field Gravity*, Ph.D. thesis at Massachusetts Institute of Technology [hdl:1721.1/77256].
3. Betancourt, M., **Stein, L. C.** (2011) *The Geometry of Hamiltonian Monte Carlo*, [arXiv:1112.4118].
2. **Stein, L. C.** (2009), *Binary Inspiral Gravitational Waves from a Post-Newtonian Expansion*, Contribution to the Wolfram Demonstrations Project, `http://demonstrations.wolfram.com/BinaryInspiralGravitationalWavesFromAPostNewtonianExpansion/`
1. **Stein, L. C.** (2006), *Gravitational Wave Burst Source Localization in a Coherent Network Analysis*, Senior thesis at California Institute of Technology

Invited Talks

52. UNC physics department colloquium — February 2024
51. UIUC astrophysics seminar — December 2023
50. Harvard CMSA GR seminar — October 2023
49. UMass Amherst, Amherst Center for Fundamental Interactions seminar — September 2023
48. Albert Einstein Institute, "Connecting the Dots" panel discussion — June 2023
47. Queen Mary Univ. of London, Gravitational memory workshop — June 2023

46. Utah State University, Theoretical Physics Talks, March 2023
45. Iowa State, Physics and astronomy department colloquium, October 2022
44. UT Austin, Weinberg Institute seminar, October 2022
43. Vanderbilt, Physics and astronomy department colloquium, September 2022
42. ICERM, Advances in CS Classical and Quantum Gravity, May 2022
41. Flatiron CCA, Ringdown workshop, invited overview talk, February 2022
40. DAMTP (University of Cambridge), HEP/GR colloquium, January 2022
39. SISSA, Current challenges in gravitational physics workshop, April 2021
38. Flatiron CCA, Gravitational wave astronomy group seminar, January 2021
37. University of Birmingham, astrophysics seminar September 2020
36. Albert Einstein Institute, ACR division seminar July 2020
35. Black Hole Perturbation Toolkit, Spring 2020 workshop May 2020
34. American Physical Society Meeting April 2020
33. UVA, physics department colloquium November 2019
32. UT Dallas, physics department colloquium October 2019
31. Northwestern University, CIERA astrophysics seminar May 2019
30. ETH-ITS Zurich, "New horizons for gravity" workshop May 2018
29. UC San Diego, astrophysics seminar March 2018
28. UC Berkeley, 4D particle physics seminar March 2018
27. Kyoto University, YKIS2018a Symposium February 2018
26. Oakland University physics seminar February 2018
25. University of Wisconsin-Milwaukee gravity seminar January 2018
24. Caltech/JPL Gravitational-Wave (CaJAGWR) seminar January 2018
23. ICN UNAM, Relativity seminar December 2017
22. University of Mississippi, Astrophysics seminar November 2017
21. University of Florida, Astrophysics seminar November 2017
20. University of Nottingham, Mathematical Physics seminar July 2017
19. Sapienza University of Rome, New Frontiers in Gravitational-Wave Astrophysics June 2017
18. Rochester Institute of Technology, CCRG seminar March 2017
17. Penn State, IGC seminar March 2017
16. University of Mississippi, Strong Gravity/Binary Dynamics workshop February/March 2017
15. SUNY Stony Brook, "The universe through gravitational waves" December 2016
14. University of Pennsylvania, New Frontiers in Gravitational Radiation workshop December 2016
13. Cambridge MA, Event Horizon Telescope collaboration meeting November/December 2016
12. Northwestern University CIERA, "Fellows at the Frontiers" August/September 2016
11. Princeton University, GR@100++ panel discussion April 2016
10. Cambridge MA, Einstein fellows symposium October 2014
9. Perimeter Institute, Strong gravity seminar October 2014
8. Cornell University, Friends of astronomy outreach event November 2013
7. Cambridge MA, Einstein fellows symposium October 2013

6. SUNY Geneseo, Physics colloquium — October 2013
5. University of Maryland, UMD gravity seminar — October 2013
4. Yale University, YCAA seminar — September 2013
3. Kyoto University, YITP long-term workshop — June 2013
2. Cambridge MA, Einstein fellows symposium — October 2012
1. Cornell University, Relativity lunch — November 2011

Contributed Talks (selected)

23. American Physical Society Meeting — April 2024
22. American Physical Society Meeting — April 2023
21. LISA Symposium XIV — July 2022
20. American Physical Society Meeting — April 2021
19. American Physical Society Meeting — April 2019
18. American Physical Society Meeting — April 2018
17. Pacific Coast Gravity Meeting — March 2017
16. American Physical Society Meeting — ~~April~~ January 2017
15. Testing Gravity 2017 — January 2017
14. 21st International meeting on GR (GR21) — July 2016
13. American Physical Society Meeting — April 2016
12. Eastern Gravity Meeting — May 2015
11. American Physical Society Meeting — April 2015
10. NEB 16 Recent developments in gravity — September 2014
9. American Physical Society Meeting — April 2014
8. XXVII Texas symposium on relativistic astrophysics — December 2013
7. 20th International meeting on GR (GR20) — July 2013
6. Eastern Gravity Meeting — June 2013
5. American Physical Society Meeting — April 2013
4. Caltech TAPIR Seminar — December 2011
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2. American Physical Society Meeting — April 2011
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